AI Job Market Intelligence

Preprocessing and Feature Engineering Documentation

salaries_enhanced.csv | 147,348 rows x 27 columns

Dataset Source	aijobs.net and foorilla.com (CC0 Public Domain)
Raw File	salaries.csv	151,445 rows x 11 columns
After Cleaning	salaries_clean_final.csv	147,348 rows x 15 columns
After Engineering	salaries_enhanced.csv	147,348 rows x 27 columns
Project	CS3012 Data Visualization	Group 8	FAST-NUCES Islamabad	Spring 2026

1. End-to-End Preprocessing Pipeline

The raw aijobs.net dataset was processed through three sequential stages before being connected to Tableau. No synthetic data was introduced at any stage. Every row in the final output file is a genuine salary submission from a real professional.

Stage	Script	Action	Result
Stage 1	`02_clean.py`	Three cleaning steps to remove noise rows	151,445 rows reduced to 147,348
Stage 2	`03_title_grouping.py`	Four readable label columns added	15 columns total
Stage 3	`05_feature_engineering.py`	Twelve analytically derived columns added	27 columns total

2. Cleaning Steps (02_clean.py)

Three sequential filters were applied to the raw dataset. The row count decreases at each step while column count remains at 11. No values were modified or imputed.

Step 1 Employment Type Filter

Technique: Boolean row filter using a single equality condition on the employment_type column.

```
df = df[df['employment_type'] == 'FT']
```

Full-time (FT) records represent 99.4 percent of the entire dataset. Part-time, contract, and freelance roles have structurally lower and more variable salary ranges that would distort all aggregated comparisons if retained alongside full-time entries. 904 rows were removed.

Step 2 Year Scope Filter

Technique: Numeric inequality filter retaining only records from 2022 onward.

```
df = df[df['work_year'] >= 2022]
```

Years 2020 and 2021 together contain only 293 rows, which is statistically insufficient to support any chart type reliably. These years are retained in the trend line visualization for historical context but excluded from all other analytical charts. 293 rows were removed.

Step 3 Outlier Removal Using Percentile Capping

Technique: Quantile-based boundary detection followed by a range filter.

```
q01 = df['salary_in_usd'].quantile(0.01)     # lower bound:  $37,974
q99 = df['salary_in_usd'].quantile(0.99)     # upper bound: $385,000
df  = df[(df['salary_in_usd'] >= q01) & (df['salary_in_usd'] <= q99)]
```

The raw salary range spanned $15,000 to $800,000. Extreme values at both ends, including a single entry at $30,000,000 and several near-zero internship wages denominated in currencies with unfavorable exchange rates, would dominate chart axes and inflate or deflate averages significantly. The 1st to 99th percentile approach preserves 98 percent of the data while removing statistically rare outliers. A total of 2,900 rows were removed.

Metric	Raw Dataset	After Cleaning	Change
Total rows	151,445	147,348	4,097 removed
Minimum salary	$15,000	$37,974	+$22,974
Maximum salary	$800,000	$385,000	-$415,000
Mean salary	$157,527	$156,237	-$1,290
Median salary	$146,100	$147,000	+$900

3. Label Columns (03_title_grouping.py)

Four columns were added to convert coded or numeric values into human-readable text suitable for Tableau chart axes, color legends, and filter controls. The row count remains unchanged at 147,348.

3.1 title_group Job Title Normalization

Technique: Keyword-based string matching applied to a lowercased copy of each raw job title.

```
def group_title(title: str) -> str:
    t = title.lower()
    if any(x in t for x in ['machine learning engineer', 'ml engineer', ...]):
        return 'ML Engineer / MLOps'
    elif any(x in t for x in ['data scientist', 'data science', ...]):
        return 'Data Scientist'
    ...

df['title_group'] = df['job_title'].apply(group_title)
```

A total of 406 unique raw title strings submitted freely by survey respondents were consolidated into 15 standardized role categories. Variant spellings such as ML Eng., Machine Learning Eng., and Machine Learning Engineer all map to ML Engineer / MLOps. The original job_title column is preserved in its entirety for use in tooltips.

title_group	Avg Salary (USD)	Record Count	Share of Data
ML Engineer / MLOps	$194,303	9,073	6.2%
Software Engineer	$180,027	21,860	14.8%
Research Scientist	$175,054	7,089	4.8%

Product Manager	$174,293	3,455	2.3%
Manager / Director	$174,149	4,092	2.8%
Data Architect	$170,929	4,525	3.1%
AI / ML Researcher	$167,597	3,306	2.2%
Data Scientist	$160,003	20,784	14.1%
Other Tech	$151,747	30,373	20.6%
Data Engineer	$147,864	16,510	11.2%
Analytics Engineer	$136,245	4,089	2.8%
Consultant / Analyst	$125,379	2,528	1.7%
Data Governance & Ops	$110,824	3,380	2.3%
Data Analyst	$107,465	14,795	10.0%
Other	$151,297	1,489	1.0%

3.2 work_mode, exp_label, size_label Code-to-Label Mapping

Technique: Dictionary mapping using the pandas .map() method applied vectorially across all rows.

```
df['work_mode']  = df['remote_ratio'].map(
    {0: 'On-site', 50: 'Hybrid', 100: 'Remote'})

df['exp_label']  = df['experience_level'].map(
    {'EN': 'Entry Level', 'MI': 'Mid Level',
     'SE': 'Senior',      'EX': 'Executive'})

df['size_label'] = df['company_size'].map(
    {'S': 'Small', 'M': 'Medium', 'L': 'Large'})
```

These three columns replace numeric or single-letter coded values with descriptive text that Tableau can display directly on chart axes and in filter controls without requiring any additional calculated fields inside the workbook.

4. Feature Engineering (05_feature_engineering.py)

Twelve new columns were derived exclusively from existing real data values using four distinct techniques. No external data was introduced and no values were synthetically generated.

Technique 1 Fixed-Width Binning

Column produced: salary_band

Definition: Assigns each numeric value to a named category based on manually specified boundary thresholds.

Function used: `pandas pd.cut()`

```
bins   = [0, 60_000, 100_000, 150_000, 200_000, 260_000, 385_001]
labels = ['<$60K', '$60K-$100K', '$100K-$150K',
          '$150K-$200K', '$200K-$260K', '$260K+']

df['salary_band'] = pd.cut(df['salary_in_usd'], bins=bins, labels=labels)
```

The bin boundaries were chosen deliberately to align with meaningful career salary thresholds: $100,000 represents the approximate entry threshold for senior individual contributor roles; $150,000 corresponds closely to the dataset median; $200,000 separates senior from executive compensation; and $260,000 captures the upper tier. Equal-width boundaries would have produced distorted buckets given the right-skewed salary distribution.

Tableau use: Color encoding on scatter and bubble charts, filter dropdown by pay tier, histogram-style distribution bar chart.

salary_band	Salary Range	Record Count	Share of Data
<$60K	$37,974 to $60,000	5,742	3.9%
$60K-$100K	$60,001 to $100,000	26,572	18.0%
$100K-$150K	$100,001 to $150,000	45,617	31.0%
$150K-$200K	$150,001 to $200,000	35,738	24.3%
$200K-$260K	$200,001 to $260,000	22,901	15.5%
$260K+	$260,001 to $385,000	10,778	7.3%

Technique 2 Quantile Binning

Column produced: salary_quartile

Definition: Divides data into equal-frequency groups by computing boundary values from the data itself, such that each group contains the same number of rows.

Function used: `pandas pd.qcut()`

```
df['salary_quartile'] = pd.qcut(
    df['salary_in_usd'],
```

```
    q=4,
    labels=['Q1 - Bottom 25%', 'Q2 - 25 to 50%',
            'Q3 - 50 to 75%',  'Q4 - Top 25%']
)
```

The key distinction between this technique and fixed-width binning is that pd.qcut does not require the analyst to specify dollar thresholds. Instead, it finds the thresholds automatically so that each quartile bucket contains approximately the same number of records, in this case roughly 36,800 rows per quartile. This is appropriate when the goal is to identify relative position within the distribution rather than classify by absolute salary level.

Tableau use: Color-encode which quartile each data point, country, or role belongs to on any chart.

salary_quartile	Computed Range	Record Count	Upper Boundary
Q1 - Bottom 25%	$37,974 to $107,711	36,837	$107,711
Q2 - 25 to 50%	$107,712 to $147,000	36,979	$147,000
Q3 - 50 to 75%	$147,001 to $196,900	36,732	$196,900
Q4 - Top 25%	$196,901 to $385,000	36,800	$385,000

Technique 3 Dictionary Lookup and Mapping

Columns produced: exp_order, region, is_us, is_ai_core

Definition: Replaces each value in a column with a corresponding value from a pre-defined key-to-value dictionary. Applied vectorially across all rows in a single operation.

Functions used: `pandas .map(dict) and .apply(lambda)`

exp_order Ordinal Encoding

```
exp_order_map = {'Entry Level': 1, 'Mid Level': 2, 'Senior': 3, 'Executive': 4}
df['exp_order'] = df['exp_label'].map(exp_order_map)
```

Without this column, Tableau sorts the experience axis alphabetically, producing the incorrect order: Entry Level, Executive, Mid Level, Senior. Dragging exp_order to the sort field in Tableau corrects this to the proper career progression sequence: Entry Level, Mid Level, Senior, Executive.

region Country to Continent Grouping

```
region_map = {
    'US': 'North America', 'CA': 'North America', 'MX': 'North America',
    'GB': 'Europe',        'DE': 'Europe',        'FR': 'Europe', ...
}
df['region'] = df['company_location'].map(region_map).fillna('Other')
```

All 87 ISO 2-letter country codes were mapped to one of six world regions. The fillna('Other') call ensures that any country code absent from the dictionary receives the label Other rather than a null value, preventing data loss in aggregations.

region	Record Count	Average Salary	Highest-paying Country

North America	139,297	$159,976	United States $161,381
Europe	6,734	$88,069	Switzerland $137,761
Asia Pacific	864	$118,899	Hong Kong $136,853
Middle East and Africa	150	$100,635	Saudi Arabia $139,999
Latin America	209	$82,382	Mexico $96,659
Other	94	$95,527	Various

is_us and is_ai_core Binary Classification Flags

```
df['is_us'] = df['company_location'].apply(
    lambda x: 'United States' if x == 'US' else 'Rest of World')

ai_core_roles = {'ML Engineer / MLOps', 'AI / ML Researcher',
                 'Data Scientist', 'Research Scientist'}

df['is_ai_core'] = df['title_group'].apply(
    lambda x: 'AI / ML Core' if x in ai_core_roles else 'Data & Tech')
```

Since the United States accounts for 90.4 percent of all records, is_us provides a clean binary dimension for US versus Rest of World salary comparison. The is_ai_core flag separates the four roles most directly associated with AI and machine learning from the broader data and technology workforce, enabling a focused salary gap analysis between the two groups.

is_us	Count	is_ai_core	Count	Avg Salary
United States	133,143	**AI / ML Core**	40,252	$171,009
Rest of World	14,205	**Data & Tech**	107,096	$150,685

Technique 4 GroupBy Aggregation with Row-Level Merge Back

Columns produced: year_avg_salary, salary_vs_year_avg, salary_vs_year_avg_pct, role_avg_salary, salary_vs_role_avg, salary_vs_role_avg_pct

Definition: Computes a summary statistic at the group level, then broadcasts that single group-level value back to every individual row belonging to that group. The per-row result is then used to compute difference and percentage deviation measures.

Functions used: `groupby().mean()`, `.map()`, arithmetic subtraction, division

Salary vs Year Average (3 columns)

```
# Step 1: compute one average per year  (4 values total)
year_avg = df.groupby('work_year')['salary_in_usd'].mean()
# Produces:  2022: $138,188    2023: $154,304
#            2024: $158,107    2025: $155,334

# Step 2: broadcast back to every row
df['year_avg_salary'] = df['work_year'].map(year_avg)

# Step 3: dollar difference
df['salary_vs_year_avg'] = df['salary_in_usd'] - df['year_avg_salary']

# Step 4: percentage difference
df['salary_vs_year_avg_pct'] = (
    df['salary_vs_year_avg'] / df['year_avg_salary'] * 100).round(1)
```

After Step 2, every row in 2025 carries the value $155,334 in the year_avg_salary column regardless of that row's actual salary. Steps 3 and 4 then compute how far each individual salary deviates from that year's average, both in absolute dollars and as a percentage. This enables Tableau to color-encode data points on a scatter chart as above or below the year benchmark.

work_year	salary_in_usd	year_avg_salary	salary_vs_year_avg	vs_year_avg_pct
2025	$348,516	$155,334	+$193,182	+124.4%
2025	$232,344	$155,334	+$77,010	+49.6%
2025	$145,400	$155,334	-$9,934	-6.4%
2024	$81,600	$158,107	-$76,507	-48.4%

Salary vs Role Average (3 columns)

```
# Step 1: compute one average per title_group  (15 values total)
role_avg = df.groupby('title_group')['salary_in_usd'].mean()

# Step 2: broadcast back to every row
df['role_avg_salary'] = df['title_group'].map(role_avg)

# Step 3 and 4: difference and percentage
df['salary_vs_role_avg']     = df['salary_in_usd'] - df['role_avg_salary']
df['salary_vs_role_avg_pct'] = (
```

```
    df['salary_vs_role_avg'] / df['role_avg_salary'] * 100).round(1)
```

The same four-step pattern is applied using title_group as the grouping key instead of work_year. Each of the 15 role categories produces one average value which is then merged back to all rows belonging to that category. The resulting deviation columns reveal, for example, that a Data Scientist earning $145,400 sits 9.1 percent below the $160,003 average for all Data Scientists in the dataset.

title_group	salary_in_usd	role_avg_salary	salary_vs_role_avg	vs_role_avg_pct
Manager / Director	$348,516	$174,149	+$174,367	+100.1%
Manager / Director	$232,344	$174,149	+$58,195	+33.4%
Data Scientist	$145,400	$160,003	-$14,603	-9.1%
Data Scientist	$81,600	$160,003	-$78,403	-49.0%
ML Engineer/MLOps	$265,000	$194,303	+$70,697	+36.4%

5. Complete Column Reference salaries_enhanced.csv

All 27 columns with data type, origin, and recommended use in Tableau. Columns 16 through 27 are feature-engineered and did not exist in the raw dataset.

No.	Column Name	Type	Origin	Recommended Use in Tableau
1	work_year	Integer	Raw	Time axis on the salary trend line chart (Sheet 1)
2	experience_level	String	Raw	Reference only. Use exp_label for display in all charts
3	employment_type	String	Raw	Constant value FT throughout. Exclude from all charts
4	job_title	String	Raw	Tooltip display only. 406 unique raw title variants
5	salary	Integer	Raw	Local currency value. Do not use for any salary analysis
6	salary_currency	String	Raw	Reference filter only. USD = 94.5 percent of records
7	salary_in_usd	Integer	Raw	Primary measure. Use on every sheet for salary analysis
8	employee_residence	String	Raw	Country of the employee. Geographic filter dimension
9	remote_ratio	Integer	Raw	0 / 50 / 100 numeric. Use work_mode for display
10	company_location	String	Raw	ISO 2-letter country code. Geographic role for map
11	company_size	String	Raw	S / M / L codes. Use size_label for display
12	title_group	String	Label	Primary role dimension. Bar chart axis (15 categories)
13	work_mode	String	Label	On-site / Hybrid / Remote. Donut chart and stacked bar
14	exp_label	String	Label	Entry Level through Executive. Experience chart axis
15	size_label	String	Label	Small / Medium / Large. Size filter and dual-axis comparison
16	salary_band	Category	Engineered	Color encoding by pay tier. Salary tier filter dropdown
17	salary_quartile	Category	Engineered	Color-encode relative position (Q1 through Q4) on any chart
18	exp_order	Integer	Engineered	Drag to Sort field to fix axis order to career progression

19	`region`	String	**Engineered**	Regional salary comparison bar chart. Six-group dimension
20	`is_us`	String	**Engineered**	United States vs Rest of World split and toggle filter
21	`is_ai_core`	String	**Engineered**	AI/ML Core vs Data and Tech salary comparison
22	`year_avg_salary`	Float	**Engineered**	Reference line value on trend chart. Tooltip context field
23	`salary_vs_year_avg`	Float	**Engineered**	Dollar deviation from year average. Tooltip detail
24	`salary_vs_year_avg_pct`	Float	**Engineered**	Percent deviation from year average. Scatter color encoding
25	`role_avg_salary`	Float	**Engineered**	Reference line per role on bar chart. Tooltip context field
26	`salary_vs_role_avg`	Float	**Engineered**	Dollar deviation from role average. Tooltip detail
27	`salary_vs_role_avg_pct`	Float	**Engineered**	Percent deviation from role average. Green/red scatter color

Color key: White / light grey = original raw columns **Blue tint** = readable label columns (Stage 2)
Green tint = feature-engineered columns (Stage 3)